Which Privacy Layer Stops Which Attack? An Attack-Grounded Comparison of Central DP, Local DP, Anonymisation and Compressive Privacy for Census-Income Prediction

Deon Ajit Menezes
Security and Privacy for Machine Learning (Honors)
Roll No. 16014223030, Batch B1

***Abstract*—Privacy-preserving machine learning offers a crowded menu of defences – central differential privacy (DP), local DP, k-anonymity, and compressive privacy – but they are almost always benchmarked in isolation, on different datasets and against different metrics, leaving the practitioner without a like-for-like answer to a simple question: for a given exposure and attack, which mechanism should I deploy, and what does it cost? We place all four families on a single pipeline with four exposure points (collection, publication, feature sharing, deployment), a common utility metric and a common attack suite (membership inference, attribute inference, linkage), using UCI Adult with an ACSIncome replication. We find that each exposure's matched mechanism measurably neutralises its attack at a modest, quantified utility cost; that output masking is a much weaker membership-inference defence than DP; and that a layered privacy-aware DCA followed by DP-SGD covers two exposures at once for essentially the single-mechanism price. The result is a *defence-coverage matrix* that makes the mismatches explicit – most sharply, that k-anonymity does nothing against membership inference. All mechanisms are implemented from scratch and the study is fully reproducible.**

***Index Terms*—differential privacy, local differential privacy, k-anonymity, compressive privacy, membership inference, privacy–utility trade-off.**

I. Introduction

A single personal record passes through several hands on its way through a machine-learning pipeline: an untrusted collector, an analyst who receives a published table, a third-party model-training service, and finally the deployed query API. Each hand-off is a distinct privacy exposure with its own canonical attack, and the privacy literature has produced a distinct defence for each: local differential privacy (LDP) for collection [1], [2], k-anonymity and its refinements for publication [3]–[5], compressive privacy for feature sharing [6], and central differential privacy (DP) for deployment [7], [8].

These mechanisms are rarely compared head-to-head because they operate at different layers and are scored on different yardsticks. This paper supplies the missing common ground: one dataset, one utility metric, one attack suite, and a resulting *coverage matrix* that states, per attack, which mechanism stops it and at what utility cost. Our contributions are: (i) a single-benchmark evaluation of four defence families against a shared attack suite; (ii) from-scratch implementations of the DP learners, six LDP oracles, Mondrian with ℓ-diversity/t-closeness, and PCA/DCA/privacy-aware DCA; (iii) a layered DCA+DP-SGD defence; and (iv) an ACSIncome replication confirming the findings are not specific to the dated Adult benchmark.

II. Threat Model and Methodology

We consider four exposure points. **P1 Collection:** an untrusted curator observes raw attributes (attribute disclosure). **P2 Publication:** a record-level table is released (linkage re-identification and homogeneity attacks). **P3 Feature sharing:** the feature matrix is sent to a third-party learner (sensitive-attribute inference). **P4 Deployment:** the trained model answers queries (membership inference). The matched defences are, respectively, LDP, k-anonymity, compressive projection, and central DP; Module-1 attacks score every configuration on identical train/test splits.

Attacks. Membership inference uses a loss-threshold attack [9] and a shadow-model attack [10], reported by AUC and TPR at 1% FPR [11]. Attribute inference trains an adversary to reconstruct `sex` from the released attributes; linkage measures quasi-identifier uniqueness.

III. Mechanisms

Central DP (M2). Laplace/Gaussian primitives; DP Naive Bayes with Laplace noise on class-conditional counts; DP logistic regression by objective perturbation; DP-SGD [8] via Opacus with an RDP accountant; and a noisy-Lloyd DP k-means. **LDP (M3).** GRR, SUE/OUE, and OLH frequency oracles; Laplace, Duchi 1-bit and the Piecewise Mechanism [12] for numeric means; and an LDP Naive Bayes classifier. **Anonymisation (M4).** Mondrian [13] with distinct-ℓ diversity and total-variation t-closeness, plus output-modification defences (label-only, rounded confidence). **Compressive privacy (M5).** PCA, Discriminant Component Analysis (DCA), and privacy-aware DCA that suppresses the sensitive label's discriminant subspace, scored by downstream utility and reconstruction error.

TABLE I
MEMBERSHIP INFERENCE PER BASELINE.

Model	Loss-MIA AUC	TPR@1%FPR	Shadow-MIA AUC
Logistic Reg.	0.497	0.010	0.502
Naive Bayes	0.498	0.439	0.501
Random Forest	0.592	0.311	0.569
MLP	0.532	0.035	0.499

IV. Experimental Setup

Datasets. UCI Adult (CC BY 4.0; 45,222 rows after cleaning; 31,656 train / 13,566 test; 103 encoded features) and, for external validity, ACSIncome (2018 ACS PUMS, California, 30,000-row subsample). **Metrics.** Utility: accuracy, balanced accuracy, F_1, ROC-AUC. Privacy: ε (at $\delta = 10^{-5}$), $k/\ell/t$, MIA AUC/advantage/TPR@1%FPR, attribute-inference accuracy, and LDP estimator MSE. Sweeps are averaged over multiple seeds.

V. Results

Membership inference. Only the overfitting Random Forest is beaten by the MIA (loss-AUC 0.592, shadow-AUC 0.569); regularised linear models sit at chance, confirming the overfitting–leakage link [9].

Central DP (Fig. 1). With class-balanced training, DP-SGD and DP-NB hold $F_1 \approx 0.61$ across $\varepsilon \in [0.5, 8]$ – within 0.02–0.05 of the non-private MLP – while the accountant's spent ε matches the target. The DP tax is small but real, and falls on unsupervised structure too (DP k-means).

LDP (Fig. 2). On `education`'s small domain OUE dominates GRR and OLH, and the Piecewise Mechanism beats Duchi/Laplace at moderate-to-high ε – the textbook orderings, reproduced.

Anonymisation and output masking. Sweeping Mondrian k removes the homogeneity leakage that k-anonymity alone leaves: hundreds of homogeneous classes at k=2 fall to zero by $k \geq 10$ as ℓ rises and t-closeness declines, at under two points of utility. Output masking only partially blunts the MIA (AUC 0.594 $\rightarrow$ 0.566), far less than DP.

Compressive privacy (Fig. 3). PCA climbs to perfect `sex` recovery; privacy-aware DCA holds income accuracy near 0.84 while keeping `sex` recovery far lower across all projection dimensions.

Coverage matrix (Table V). Consolidating the above answers the title question and exposes the mismatches, most sharply that k-anonymity offers no membership-inference protection.

External validity. On ACSIncome the Random Forest again leaks membership (AUC $>$ 0.69) and DP-SGD trades a few F_1 points for protection, so the conclusions are not an artefact of Adult.

VI. Discussion and Limitations

The study uses two US-census datasets and a qualitative top-level matrix over quantitative per-cell measurements; the from-scratch DP-LR uses output perturbation (we report the more robust `diffprivlib` curve), and t-closeness is a total-variation surrogate. None of these reverse a result. The central lesson is that *privacy is layer-specific*: a mechanism that is excellent at its own exposure (e.g. k-anonymity for publication) can be useless at another (membership inference), which is precisely the kind of mistake a coverage matrix prevents.

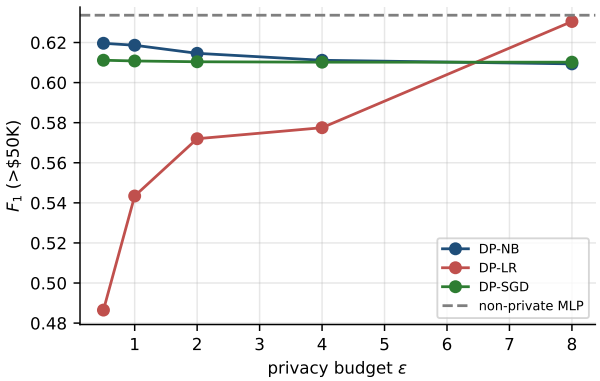


Fig. 1. Central-DP learners: F_1 versus privacy budget ε.

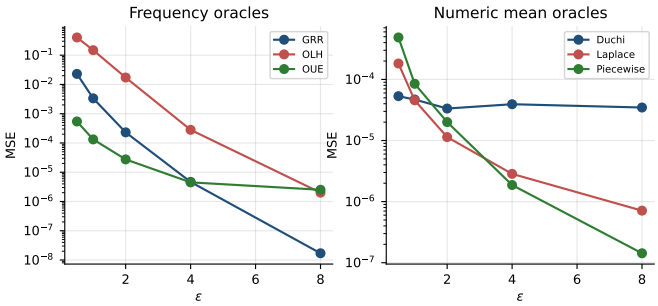


Fig. 2. LDP estimation error versus ε: frequency oracles (left), numeric mean oracles (right).

VII. Conclusion

A single-benchmark, attack-grounded comparison shows each privacy family earning its place at its own exposure point at a modest, measured utility cost, while a layered DCA+DP-SGD pipeline covers two exposures at once. The coverage matrix turns a crowded menu into an actionable map. Future work extends the matrix to federated and cryptographic defences and to non-census modalities.

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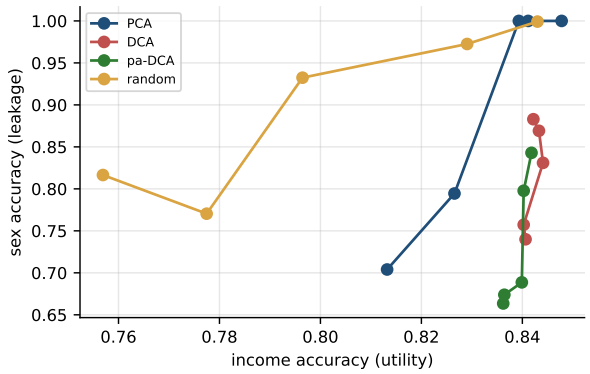


Fig. 3. Utility versus sensitive-leakage frontier for the compressive projections ($p = 2 \ldots 40$).

TABLE II
DP SUPERVISED F_1 BY ε.

ε	DP-NB F_1	DP-LR F_1	DP-SGD F_1
0.5	0.620	0.486	0.611
1.0	0.619	0.543	0.611
2.0	0.615	0.572	0.610
4.0	0.611	0.577	0.610
8.0	0.609	0.630	0.610

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TABLE III
COMPRESSIVE PROJECTIONS AT p=10: UTILITY (INCOME) VS LEAKAGE (SEX).

Projection	p	Income acc.	Sex acc.	Recon. err.
PCA	10	0.839	1.000	0.492
DCA	10	0.844	0.831	0.751
pa-DCA	10	0.840	0.689	0.780
random	10	0.796	0.932	0.689

TABLE IV
ACSINCOME (CALIFORNIA, 2018) REPLICATION.

Setting (ACSIncome, CA 2018)	F_1	MIA AUC
baseline RF	0.762	0.693
DP-SGD eps=4	0.732	–

TABLE V

DEFENCE-COVERAGE MATRIX (✓ STOPS, ∼ PARTIAL, × NO EFFECT) – THE HEADLINE RESULT CONSOLIDATING EVERY MEASUREMENT.

Attack	Mechanism	Stops?	Why
Membership inference	DP-SGD / DP-NB	✓	Central DP bounds per-record influence
Membership inference	Output modification	✓	Removes the confidence signal
Attribute inference (sex)	Privacy-aware DCA	✓	Suppresses the sensitive discriminant
Linkage / re-identification	k-anonymity (Mondrian)	✓	Generalises quasi-identifiers
Homogeneity disclosure	ℓ-diversity / t-closeness	✓	Diversifies each class
Collection-time disclosure	Local DP	✓	No trusted curator needed
Membership inference	k-anonymity	×	Protects the table, not the model
Attribute inference	Central DP alone	∼	Partial; not targeted at the sensitive attr.

TABLE VI

MONDRIAN k-ANONYMITY SWEEP: AS k GROWS THE HOMOGENEITY LEAKAGE THAT k-ANONYMITY ALONE LEAVES IS REMOVED (ℓ RISES, t-CLOSENESS FALLS, DISCLOSURE $\to 0$) AT UNDER TWO POINTS OF UTILITY.

Target k	Ach. k	Min ℓ	Homog.	Disclosed	t	Util. acc
2	2	1	339	0.019	0.995	0.842
5	5	1	5	0.001	0.889	0.839
10	10	2	0	0.000	0.812	0.837
20	20	3	0	0.000	0.700	0.837
50	51	9	0	0.000	0.521	0.847